

Project Proposal: AI Exam Prep

Many students struggle to prepare effectively for exams. Common study methods such as reading notes, watching lectures, or creating your own practice question are time consuming and do not provide enough personalized practice or timely feedback. Students often do not know which topics they are weak in until after taking an exam, when it is too late to improve. Tutors and instructors can offer personalized feedback, but they are expensive, time-consuming, and not available to every student. Existing online platforms often rely on fixed workflows and predefined questions, which limit adaptability.

The real-world problem being addressed is the lack of scalable, affordable, and personalized exam preparation tools. Schools, online learning platforms, and individual learners all need systems that can imitate exam conditions, evaluate performance, and guide improvement without requiring manual grading or tutoring. This problem is especially relevant in large classes, online education, and self-study environments, where instructors cannot provide detailed feedback to every student.

The target users for this system include university and high school students preparing for exams, self-learners studying for standardized or certification exams, educators who want automated tools to support assessments. These users need practice exams, quick feedback, and guidance on what to study next. An agentic AI solution is appropriate because exam preparation is not a single task but a sequence of interconnected tasks. These include generating questions, evaluating answers, identifying weaknesses, and adjusting future exams accordingly. A traditional system with a fixed workflow would struggle to adapt to different users and learning progress making this system a more flexible and scalable option.

An example of a scenario is a student wanting to practice for an upcoming exam. The user uploads or pastes current course materials such as lecture slides, textbook information, or notes. The system analyzes the content provided and generates a practice exam that closely matches the course topics

and difficulty level. After the student submits answers, the system grades the exam and provides feedback for each question.

Another example of a use case is a student wanting targeted practice based on weak topics. After the user uploads course materials and completes the initial practice exam, based on the results, the system identifies weak areas within the content. It then generates a shorter, focused practice exam using the same materials and gives detailed explanations for incorrect answers, helping the student improve on specific topics.

Preliminary list of agents and roles:

- Exam Generator Agent: Creates practice exam questions based on topic and difficulty
- Grading Agent: Evaluates student answers and assigns scores
- Feedback Agent: Explains mistakes and suggests improvements
- Planner Agent: Coordinates the workflow between generation, grading, and feedback
- Reviewer Agent: Checks feedback quality and consistency

Expected tools and data sources:

- Large language model API for question generation and feedback
- Local or predefined topic datasets
- Simple database or local files to store user performance
- Web-based front-end interface for user interaction

Potential challenges:

- Hallucinated or incorrect answers in generated questions or feedback
- Inconsistent grading for open-ended questions
- Managing context length for long exams or histories

Constraints:

- API rate limits and computation costs
- Privacy concerns when storing student data